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Reference Formulas

Probability and Gaussians

Mean and covariance of an affine map
$\mathbb{E}[AX+b] = A\mu + b, \quad \mathrm{Cov} = A\Sigma A^\top$

Gaussian noising, independent terms
$X_t = \alpha_t X_0 + \sigma_t \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, I)$

$\mathrm{Cov}(X_t) = \alpha_t^2 \,\mathrm{Cov}(X_0) + \sigma_t^2 I$

Score of a Gaussian
$\nabla_x \log \mathcal{N}(x; m, \sigma^2 I) = -\dfrac{x-m}{\sigma^2}$

Bayes' rule
$p(x \mid y) = \dfrac{p(y \mid x)\, p(x)}{p(y)}$

Total probability and posterior weight
$p(x) = \displaystyle\int p(x \mid y) p(y)\, dy, \;\; w(y \mid x) = \frac{p(x \mid y) p(y)}{p(x)}$

Linear algebra

Diagonalization, matrix exponential and its determinant
$A = V \Lambda V^{-1}, \qquad e^{tA} = V e^{t\Lambda} V^{-1}$

$\det e^{hM} = \prod_i e^{h m_i} = e^{h\,\mathrm{tr} M}$

Divergence is a Jacobian trace
$\nabla \cdot v = \mathrm{tr}\, J_v = \displaystyle\sum_i \partial_i v_i$

Laplacian
$\nabla^2 f = \nabla \cdot \nabla f = \displaystyle\sum_i \partial_i^2 f$

Change of variables

Pushforward
$\mathbb{P}(Y \in B) = \mathbb{P}\big(X \in \phi^{-1}(B)\big)$

Density under an invertible map
$p_Y(\phi(x)) = \dfrac{p_X(x)}{|\det J_\phi(x)|}$

Log-density form
$\log p_Y(\phi(x)) = \log p_X(x) - \log |\det J_\phi(x)|$

Flows and densities

Trajectory and flow map
$\dfrac{dx_t}{dt} = v_t(x_t), \qquad x_t = \phi_t(x_0)$

Euler step
$\hat{x}_{k+1} = \hat{x}_k + \Delta t\, v_{t_k}(\hat{x}_k)$

Continuity equation and flux
$\partial_t p_t + \nabla \cdot (p_t v_t) = 0, \quad j_t = p_t v_t$

Score identity
$s_t = \nabla \log p_t \iff p_t\, s_t = \nabla p_t$

Log density along a trajectory
$\frac{d}{dt} \log p_t(x_t) = -\nabla \cdot v_t(x_t)$

$\log p_1(x_1) = \log p_0(x_0) - \displaystyle\int_0^1 \nabla \cdot v_t \, dt$

Flow matching

Linear interpolant and target
$X_t = (1-t)X_0 + tX_1, \qquad U = X_1 - X_0$

General schedule
$X_t = \alpha_t X_0 + \beta_t X_1, \;\; U = \dot{\alpha}_t X_0 + \dot{\beta}_t X_1$

Marginal velocity
$v_t(x) = \mathbb{E}[\, U_t \mid X_t = x\,]$

Least squares gives the conditional mean
$\arg\min_a \mathbb{E}\big[\|a - U\|^2 \mid X_t = x\big] = \mathbb{E}[U \mid X_t = x]$

Conditional flow-matching equivalence
$\mathcal{L}_{\mathrm{CFM}} = \mathcal{L}_{\mathrm{FM}} + \mathrm{const}, \;\; \nabla_\theta \mathcal{L}_{\mathrm{CFM}} = \nabla_\theta \mathcal{L}_{\mathrm{FM}}$

Stochastic dynamics

SDE and Euler–Maruyama step
$dX_t = f(X_t, t)\, dt + g(t)\, dW_t$

$x_{k+1} = x_k + fh + g\sqrt{h}\, \varepsilon_k$

Itô's lemma
$d\varphi(X_t) = \Big[b \cdot \nabla\varphi + \tfrac{1}{2}\mathrm{tr}\big(\sigma\sigma^\top \nabla^2 \varphi\big)\Big] dt + \nabla\varphi^\top \sigma\, dW_t$

Fokker–Planck equation
$\partial_t p_t = -\nabla \cdot (f\, p_t) + \tfrac{1}{2} g^2 \nabla^2 p_t$

Heat equation
$\partial_t p_t = D\, \nabla^2 p_t$

Ornstein–Uhlenbeck process
$dX_t = -\kappa X_t\, dt + g\, dW_t$

$X_t \mid x_0 \sim \mathcal{N}\Big(e^{-\kappa t} x_0,\; \tfrac{g^2}{2\kappa}\big(1 - e^{-2\kappa t}\big) I\Big)$

Diffusion

Variance-preserving corruption
$X_r = \alpha_r X_0 + \sigma_r \varepsilon, \qquad \alpha_r^2 + \sigma_r^2 = 1$

Score, noise and clean estimate
$s = -\dfrac{\varepsilon_\theta}{\sigma_r}, \qquad \hat{x}_0 = \dfrac{x + \sigma_r^2 s}{\alpha_r}$

Score matching by parts
$J(\psi) = \mathbb{E}_p\big[\tfrac{1}{2}\|s_\psi\|^2 + \nabla \cdot s_\psi\big] + C$

Reverse-time SDE
$dX_t = \big[f - g^2 s_t\big] dt + g\, dW_t, \quad dt < 0$

Probability-flow ODE
$\dfrac{dX_t}{dt} = f(X_t, t) - \tfrac{1}{2} g(t)^2 s_t(X_t)$

DDPM forward chain
$x_k = \sqrt{\bar{\alpha}_k}\, x_0 + \sqrt{1 - \bar{\alpha}_k}\, \varepsilon, \;\; \bar{\alpha}_k = \displaystyle\prod_{i \le k} \alpha_i$

DDPM reverse posterior
$\tilde{\mu}_k = \dfrac{\sqrt{\bar{\alpha}_{k-1}}\, \beta_k\, x_0 + \sqrt{\alpha_k}\,(1 - \bar{\alpha}_{k-1})\, x_k}{1 - \bar{\alpha}_k}$

$\tilde{\beta}_k = \dfrac{\beta_k (1 - \bar{\alpha}_{k-1})}{1 - \bar{\alpha}_k}$

Guidance

Conditional score
$\nabla \log p(x \mid c) = \nabla \log p(x) + \nabla \log p(c \mid x)$

Guidance scale and tilted target
$s_t(x) + \gamma \nabla \log p_t(c \mid x)$

$\iff p_t^\gamma(x) \propto p_t(x) \cdot \big[p_t(c \mid x)\big]^\gamma$

Reward guidance
$s^{(R)} = s + \eta \nabla R, \qquad p^{(R)} \propto p\, e^{\eta R}$

Where the guidance term enters
SDE drift : $f - g^2\big(s + \gamma \nabla \log \ell\big)$

ODE velocity : $f - \tfrac{1}{2} g^2 \big(s + \gamma \nabla \log \ell\big)$

Classifier-free guidance
$F_{\mathrm{CFG}} = F_\varnothing + w\,(F_c - F_\varnothing)$

Multi-objective weighting
$\tilde{R}_m = \dfrac{R_m - \mu_m}{\tau_m}, \qquad R_\lambda = \displaystyle\sum_m \lambda_m \tilde{R}_m$

Information measures

Entropy and differential entropy
$H(p) = -\displaystyle\int p \log p, \qquad H(p, q) = -\int p \log q$

KL divergence and its decomposition
$D_{\mathrm{KL}}(p \| q) = \mathbb{E}_p\Big[\log \tfrac{p}{q}\Big] \;\ge 0$

$H(p, q) = H(p) + D_{\mathrm{KL}}(p \| q)$

Notation Reference

Times and clocks

Symbol	Meaning
t	Generation time: noise at $t = 0$, data at $t = 1$
r	Noising clock: clean data at $r = 0$, noise at $r = T$
$\tau = T - r$	Generation clock for the diffusion model
s, t	Two times of one interval, $s < t$
h	Step size of a numerical update

Context-dependent symbols

Symbol	Meaning
τ_1, τ_2	Positive reward scales. *Not* a time
c	Strength of the score correction, a real constant
$c_{s,t}$	Noise coefficient of the interval update
c (as a condition)	Class label, as in p_t^c and ℓ_r^c
α_j, $\bar{\alpha}_j$	DDPM schedule and its running product
α	Weight between the two objectives, $0 < \alpha < 1$
μ, Σ	Mean and covariance of the data sample D
$\widetilde{\mu}_k$	DDPM reverse posterior mean
μ_1, μ_2	Reference means of the two rewards

Operators

Symbol	Meaning
∇	Gradient in x
$\nabla\cdot$	Divergence in x, equal to tr of the Jacobian
Δ, ∇^2	Laplacian, $\nabla\cdot\nabla$. The two are the same here
$\|\cdot\|$	Euclidean norm

Densities and fields

Symbol	Meaning
p_t, $q_t(\cdot \mid D)$	Target marginal; conditional given the endpoint
ρ_t	The density the sampler actually has
π_t	Guided density
v_t, s_t, b_t	Target velocity, target score, sampler drift
v_θ, s_ψ	The learned velocity and score
U_t, G_t	Conditional velocity and score targets
λ, $a(t)$	Noise strength and its time dependence
w, η	Classifier-free and reward guidance strengths
κ, ν	Ablation switches, each 0 or 1
ζ	Unit-rescaling factor, $\zeta > 1$